Physics

Quarter 1 Textbook

Grade 97

Scientific Inquiry & Kinematics

Name: _____ Class: _____ Date: _____

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Unit 1 — Scientific Inquiry

Scientific Inquiry — Observations

Lesson 1 of 6

Distinguish qualitative from quantitative observations and objective from subjective statements, and explain what makes a statement scientific.

Observations

> Note
> An **observation** is the act of gathering information using your senses or instruments. Information gathered from an observation is given as a **statement**.

Qualitative and Quantitative Statements

> Note
> **Qualitative**: descriptive, non-numerical data. **Quantitative**: numerical, measured data.

Examples

Qualitative	Quantitative
the solution is blue	the solution has volume $50.0\ \mathrm{mL}$
the object is hot	the temperature is $80^{\circ}\mathrm{C}$
the surface is rough	the roughness average is $2.3\ \mu\mathrm{m}$
the sound is loud	the sound intensity is $85\ \mathrm{dB}$

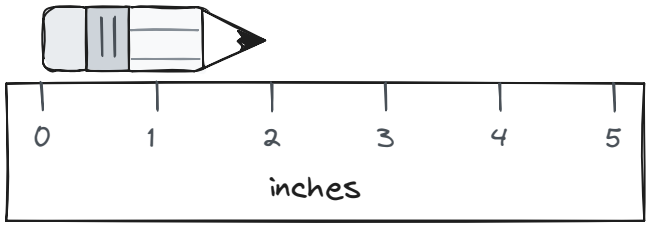
Qualitative vs Quantitative



Make a qualitative statement about Pikachu.

Make a quantitative statement about Pikachu.

Pencil Example



Qualitatively, this pencil is small. Quantitatively, this pencil measures $2\ \text{in}$.

Subjective and Objective Statements

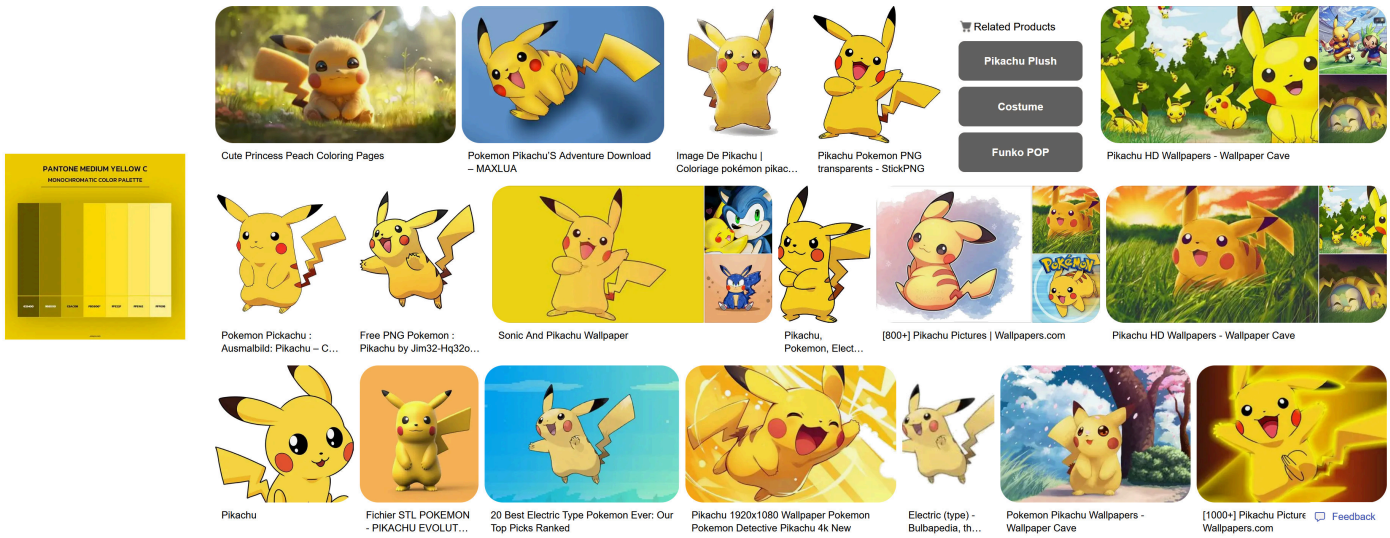
> Note
> **Subjective**: opinion-based statements that vary by person. **Objective**: fact-based, verifiable statements.

Examples

Subjective	Objective
the soup tastes salty	the soup contains $800\ \text{mg}$ of sodium
the room feels warm	the room temperature is $24°\text{C}$
chocolate is the best flavor	chocolate is the most-sold flavor
Pikachu is cute	Pikachu is yellow

Combined Statements

Statements can be qualitative **and** objective. Pikachu **is yellow**.



What makes a statement *scientific*?

> Important
> A **scientific statement** must be **objective**, **testable**, and **falsifiable**.
> - **Testable**: data can be gathered through observation or experiment.
> - **Falsifiable**: it is *possible* to be proven false through testing.

Examples

Non-scientific	Scientific
The red car is fast.	The red car has a top speed of $210\ \frac{km}{h}$ (quantitative)
The water feels hot.	The water is boiling and producing steam. (qualitative)
This is a heavy backpack.	The backpack has a mass of $12\ \text{kg}$. (quantitative)
The flower is pretty.	The flower has pink petals. (qualitative)
Unicorns do not exist.	The data does not support the existence of a unicorn. (qualitative)

Scientific Inquiry — Measurement & Metric Prefixes

Lesson 2 of 6

Measure with the correct tools, distinguish accuracy from precision, and convert between metric prefixes and scientific notation.

What Is a Measurement?

> Note
> A **measurement** is an observation that compares an unknown quantity to a known quantity.

Tool	Quantity Measured	SI Units
Meter stick	distance, displacement	meter, m
Stopwatch	time	second, s
Mass scale	mass	kilogram, kg
Force scale	force	Newton, N

Measuring Tools



Accuracy

- How close the average of the measurements is to the actual value.

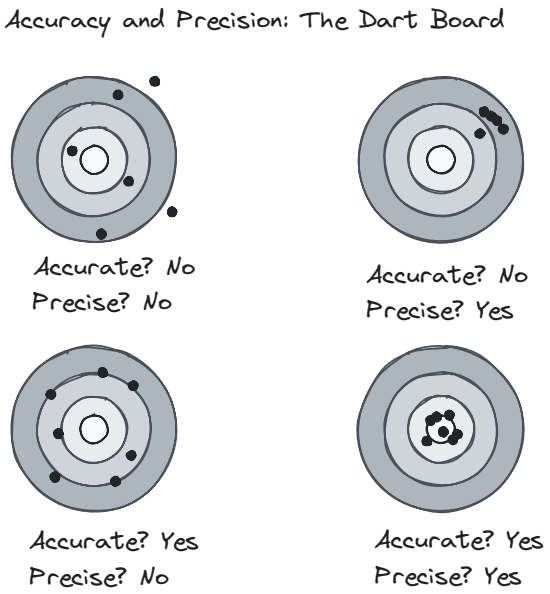
Lower Accuracy (Actual: $30\ \mathrm{kg}$**)**	**Higher Accuracy (Actual:** $30\ \mathrm{kg}$**)**
$29\ \mathrm{kg}$	$29\ \mathrm{kg}$
$30\ \mathrm{kg}$	$29\ \mathrm{kg}$
$28\ \mathrm{kg}$	$30\ \mathrm{kg}$
$28\ \mathrm{kg}$	$31\ \mathrm{kg}$
Average: $28.8\ \mathrm{kg}$	**Average:** $29.8\ \mathrm{kg}$

Precision

- How close together the measurements are to each other.

Lower Precision	Higher Precision
$50\ \mathrm{mm}$	$54\ \mathrm{mm}$
$54\ \mathrm{mm}$	$56\ \mathrm{mm}$
$59\ \mathrm{mm}$	$55\ \mathrm{mm}$
$56\ \mathrm{mm}$	$56\ \mathrm{mm}$
Range: $9\ \mathrm{mm}$	**Range:** $2\ \mathrm{mm}$

Accuracy vs Precision



Metric System

> **Important**
> The **metric system** is a decimal-based system of measurement used in science — built on powers of 10 and used by scientists worldwide.

Why Prefixes?

> **Tip**
> Prefixes make very large or very small numbers easier to report.

> The wavelength is $0.000000673\ \text{m}$

> The wavelength is $673\ \text{nm}$

- Prefixes express very big or very small numbers.
- My phone's storage is $512\ \text{GB}$: $1\ \text{GB} = 10^9\ \text{bytes}$.

Metric Prefixes

Prefix	Symbol	Order of Magnitude
Tera-	T	10^{12}
Giga-	G	10^{9}
Mega-	M	10^{6}
Kilo-	**K**	10^{3}
hecto-	**h**	10^{2}
deka-	**da**	10^{1}
(none)	**(base)**	$10^{0} = 1$
deci-	**d**	10^{-1}
centi-	**c**	10^{-2}
milli-	**m**	10^{-3}
micro-	μ	10^{-6}
nano-	n	10^{-9}
pico-	p	10^{-12}

Scientific Notation

Scientific notation writes a number as $a \times 10^n$

Number	Scientific Notation
$602{,}000{,}000{,}000{,}000{,}000{,}000$	6.02×10^{23}
0.000000000001	1.0×10^{-12}

- The exponent n moves the decimal point.
- A metric prefix is just a scientific-notation scale.
  - (e.g. kilo $= 10^3$, nano $= 10^{-9}$)

Very Big Numbers → Scientific Notation

Move the decimal point left; the number of places moved is the positive exponent.

$$9{,}300{,}000 = 9.3 \times 10^6$$

$$150{,}000{,}000 = 1.5 \times 10^8$$

Very Small Numbers → Scientific Notation

Move the decimal point right; the number of places moved is the negative exponent.

$$0.0000045 = 4.5 \times 10^{-6}$$

$$0.000000025 = 2.5 \times 10^{-8}$$

Big Prefixes → Scientific Notation

The prefix gives the power of ten, so a measurement becomes $a \times 10^n$.

$$3.0\ \text{Gm} = 3.0 \times 10^9\ \text{m}$$

$$2.5\ \text{Tm} = 2.5 \times 10^{12}\ \text{m}$$

Small Prefixes → Scientific Notation

Small prefixes give negative powers of ten.

$$4.2\ \mu\text{s} = 4.2 \times 10^{-6}\ \text{s}$$

$$750\ \text{nm} = 7.5 \times 10^{-7}\ \text{m}$$

Converting Prefixes Examples

5 kilometers to meters

$$5\ \cancel{\text{km}} \left(\frac{1000\ \text{m}}{1\ \cancel{\text{km}}} \right) = 5000\ \text{m}$$

15 kilograms to milligrams

$$15\ \cancel{\text{kg}} \left(\frac{1000\ \cancel{\text{g}}}{1\ \cancel{\text{kg}}} \right) \left(\frac{1000\ \text{mg}}{1\ \cancel{\text{g}}} \right) = 15{,}000{,}000\ \text{mg}$$

2.5 hours to seconds

$$2.5\ \cancel{\text{hr}} \left(\frac{60\ \cancel{\text{min}}}{1\ \cancel{\text{hr}}} \right) \left(\frac{60\ \text{s}}{1\ \cancel{\text{min}}} \right) = 9000\ \text{s}$$

40 km/h to m/s

$40\ \frac{\cancel{\text{km}}}{\cancel{\text{h}}} \left(\frac{1000\ \text{m}}{1\ \cancel{\text{km}}} \right) \left(\frac{1\ \cancel{\text{h}}}{3600\ \text{s}} \right) = \frac{40 \times 1000}{3600}\ \frac{\text{m}}{\text{s}} = 11.11\ \frac{\text{m}}{\text{s}}$

Scientific Inquiry — Base Units & Derived Units

Lesson 3 of 6

Distinguish base units from derived units and derive the unit of a quantity from its formula.

International System of Units (SI)

> Note
> There are **seven** SI base units.

Quantity	Unit	Symbol
length or distance	meter	m
mass	kilogram	kg
time	second	s
electric current	ampere	A
temperature	Kelvin	K
amount of substance	mole	mol
luminous intensity	candela	cd

Why Do We Need Other Units?

Imagine describing a car's speed like this:

> "The car travels 100 meters every 5 seconds."

Instead, we can express this with a **derived unit**: **meters per second** $\left(\frac{\mathrm{m}}{\mathrm{s}}\right)$.

> Important
> A **derived unit** is a combination of base units.

How Derived Units Work

- A derived unit comes from the **formula** that defines the quantity.

Example: Area

1. **Formula:** $\mathrm{Area} = \mathrm{length} \times \mathrm{width}$
2. **Base Units:** Both length and width are measured in **meters** (m) .
3. **Derived Unit:** $\mathrm{m} \times \mathrm{m} = \mathrm{m}^2$ (meters squared).

To find a derived unit:

> Tip
> 1. **Find the formula** for the quantity.
> 2. **Replace** each quantity with its **SI base unit**.
> 3. **Simplify** the expression.

Example 1: Velocity

- The formula for **average velocity** is:

$$\bar{v} = \frac{\Delta x}{t}$$

$$\bar{v} = \frac{\Delta x}{t}$$

- Δx, displacement, has the unit **meters** (m)
- t, time, has the unit **seconds** (s) .

$$\bar{v} = \frac{\mathrm{m}}{\mathrm{s}}$$

$$\bar{v} = \frac{\mathrm{m}}{\mathrm{s}}$$

- *meters per second*

Example 2: Acceleration

- The formula for **acceleration** is:

$$a = \frac{\Delta v}{t}$$

$$a = \frac{\Delta v}{t}$$

- Δv, change in velocity, has the unit $\frac{\mathrm{m}}{\mathrm{s}}$.
- t has the unit s.

$$a = \frac{\frac{\mathrm{m}}{\mathrm{s}}}{\mathrm{s}} = \frac{\mathrm{m}}{\mathrm{s}^2}$$

$$a = \frac{\frac{\mathrm{m}}{\mathrm{s}}}{\mathrm{s}} = \frac{\mathrm{m}}{\mathrm{s}^2}$$

- *meters per second squared*

Example 3: Force

- The formula for **force** is:

$$F = m \times a$$

$$F = m \times a$$

- m has the unit kg
- a has the unit $\frac{\mathrm{m}}{\mathrm{s}^2}$.

$$F = \mathrm{kg}\frac{\mathrm{m}}{\mathrm{s}^2} = \mathrm{N}$$

$$F = \mathrm{kg}\frac{\mathrm{m}}{\mathrm{s}^2} = \mathrm{N}$$

- *kilograms meters per second squared*, *Newton*

Special Derived Units

Some derived units appear so often that they get their own names.

Derived Quantity	Formula	Base Units	Special Name & Symbol
Force	$F = ma$	$\frac{\mathrm{kg}\cdot\mathrm{m}}{\mathrm{s}^2}$	**Newton** (N)
Energy/Work	$W = Fd$	$\frac{\mathrm{kg}\cdot\mathrm{m}^2}{\mathrm{s}^2}$	**Joule** (J)
Power	$P = \frac{W}{t}$	$\frac{\mathrm{kg}\cdot\mathrm{m}^2}{\mathrm{s}^3}$	**Watt** (W)
Frequency	$f = \frac{1}{T}$	$\frac{1}{\mathrm{s}}$	**Hertz** (Hz)
Charge	$Q = I \cdot t$	$\mathrm{A}\cdot\mathrm{s}$	**Coulomb** (C)

Scientific Inquiry – Problem Solving

Lesson 4 of 6

Classify quantities as scalars or vectors and explain the steps used to solve a physics problem.

Scalars and Vectors

> Note
> **Scalar** — a number with units (e.g. a train's **speed** of $90\frac{\text{km}}{\text{h}}$). **Vector** — a number with units *and direction* (e.g. a train's **velocity** of $90\frac{\text{km}}{\text{h}}$ *east*).

Vector Directions

A vector can be in any direction, but we will use vectors aligned with the x and y axes.

List of Scalars and Vectors

Scalar Quantities	Vector Quantities
distance	displacement
speed	velocity
energy/work	acceleration
charge	force
	momentum

Problem Solving Steps

> Important
> 1. **What am I given? What am I finding?**
> 2. **What equation connects them?**
> 3. **Plug in and solve.**
> 4. **Check the units and direction (if applicable).**

Example 1

A train travels $180\ \text{km}$ in $2\ \text{h}$. What is its speed?

1. List the quantities.

$d = 180\ \text{km}$, $t = 2\ \text{h}$, $s = ?$

2. Use an equation that connect the quantities.

$$s = \frac{d}{t}$$

3. Plug in and solve.

$$s = \frac{180\ \text{km}}{2\ \text{h}} = 90\frac{\text{km}}{\text{h}}$$

4. Check the units and direction.

- The unit $\frac{\text{km}}{\text{h}}$ is valid for speed.
- Speed is a scalar so direction is not required.

Example 2

A bird flies $500\ \text{m}$ east in $25\ \text{s}$. What is its velocity?

1. List the quantities.

$\Delta x = 500\ \text{m}$ (east), $t = 25\ \text{s}$, $v = ?$

2. Use an equation that connect the quantities.

$$v = \frac{\Delta x}{t}$$

3. Plug in and solve.

$$v = \frac{500\ \text{m}}{25\ \text{s}} = 20\frac{\text{m}}{\text{s}}\ \text{east}$$

4. Check the units and direction.

- The unit $\frac{\text{m}}{\text{s}}$ is valid for velocity.
- Velocity is a vector and its direction is *east*.

Example 3

A car goes from $10\frac{\text{m}}{\text{s}}$ to $30\frac{\text{m}}{\text{s}}$ in 5 s. What is its acceleration?

1. List the quantities.

$v_0 = 10\frac{\text{m}}{\text{s}}$, $v = 30\frac{\text{m}}{\text{s}}$, $t = 5$ s, $a = ?$

2. Use an equation that connect the quantities.

$$a = \frac{v - v_0}{t}$$

3. Plug in and solve.

$$a = \frac{30\frac{\text{m}}{\text{s}} - 10\frac{\text{m}}{\text{s}}}{5\ \text{s}} = \frac{20\frac{\text{m}}{\text{s}}}{5\ \text{s}} = +4\frac{\text{m}}{\text{s}^2}$$

4. Check the units and direction.

The unit $\frac{\text{m}}{\text{s}^2}$ is valid for acceleration. Acceleration is a vector, but the problem did not give a direction, so we fallback to positive/negative.

Scientific Inquiry – Graphs

Lesson 5 of 6

Build and read graphs with SULTAN, find the line of best fit, y-intercept, and slope, and extrapolate from a graph.

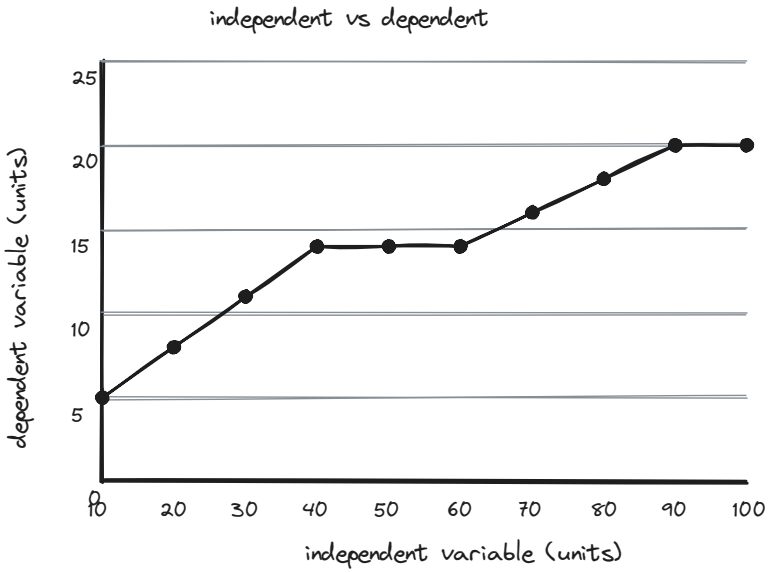
Tables

- The independent variable goes in the first column.
- **average (mean)**: add all the values, then divide by how many there are.

Name of Pet	Species	Color of Eyes	Mass (kg)
Toby	Cat	Yellow	4.2
Sugar	Dog	Brown	15.1
Hiro	Dog	Dark Brown	23.8
Tiger	Cat	Yellow	6.0
Booboo	Cat	Brown	3.5

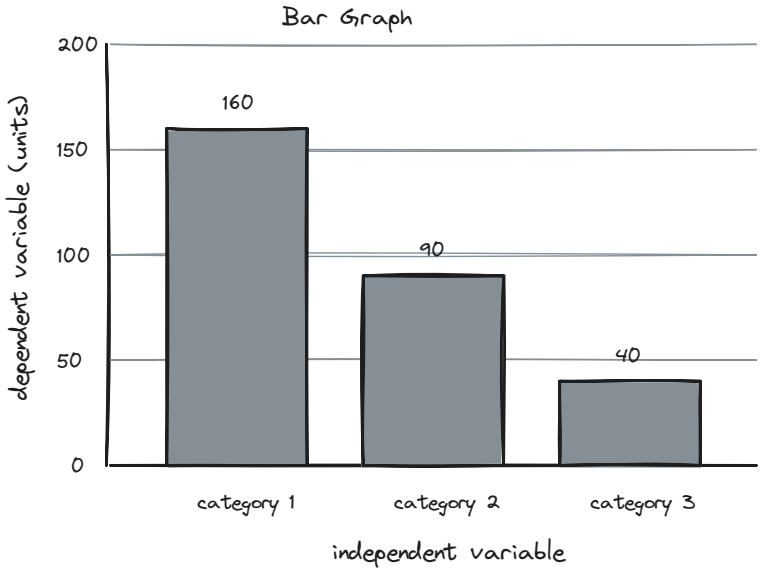
Line Graph

- Used for continuous data.



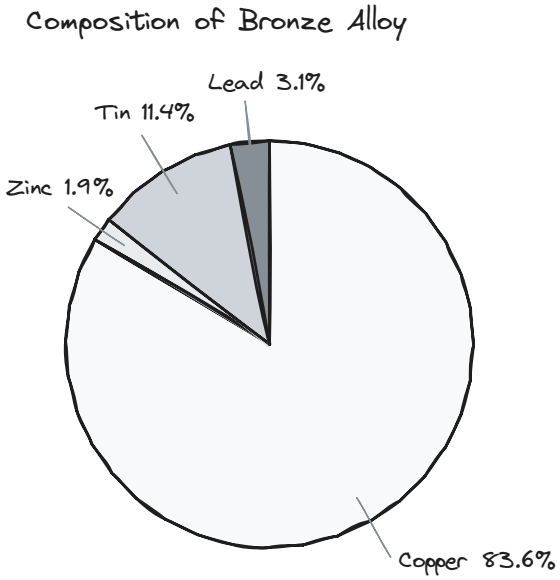
Bar Graph & Pie Chart

- Used for categories and comparisons.



Pie Chart

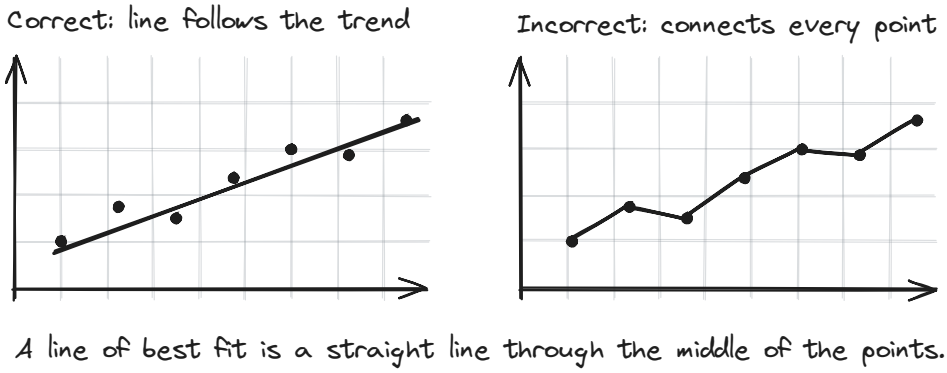
- Shows parts of a whole (percentages).



SULTAN

> **Tip**
> **SULTAN** — the rules for a good graph:
>
> - **S**cale: Choose appropriate axis scales
> - **U**nits: Label both axes with units
> - **L**egend: Include if multiple data sets
> - **T**itle: Describe what the graph shows
> - **A**xes: Label independent (x) and dependent (y)
> - **N**eat: Plot points clearly

Line of Best Fit

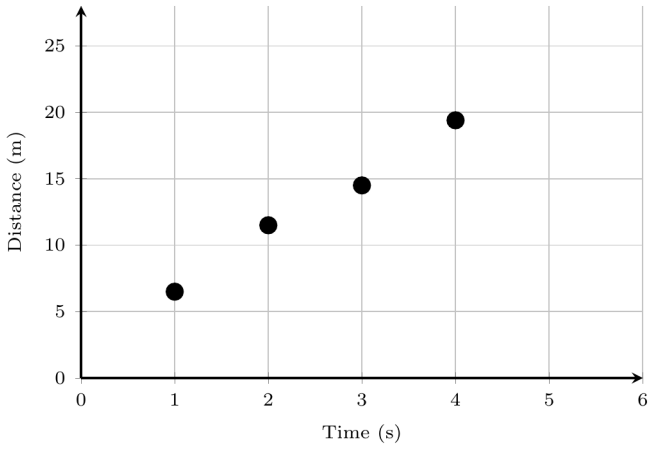


> **Warning**
> Do **not** connect the points one by one — that is a *line graph*, not a **line of best fit**. Draw one straight line that follows the trend.

- Correct: a **straight** line following the data trend
- Incorrect: a line connecting every point (this would be a *line graph*, not a *line of best fit*)

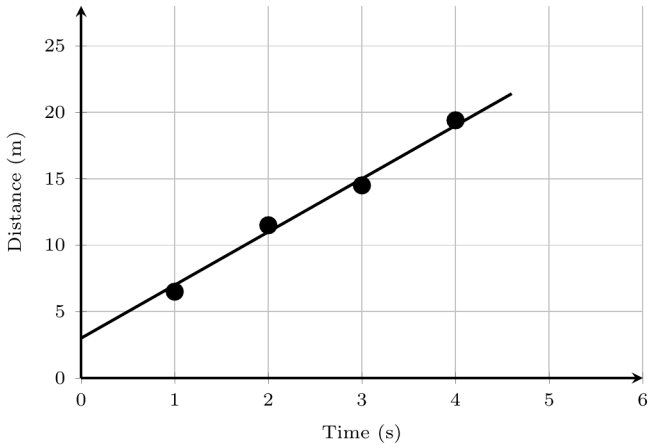
Step 1: Plot the Data

- Put the independent variable on the x-axis and the dependent variable on the y-axis.



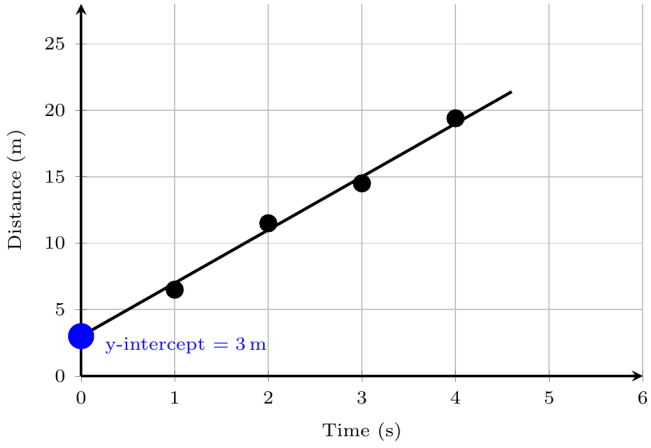
Step 2: Line of Best Fit

- Draw one straight line that follows the trend of the points.
- Don't just connect the points one by one.



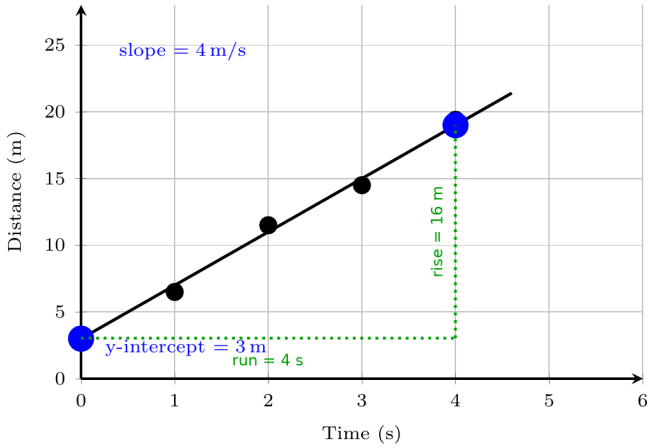
Step 3: Identify the y-Intercept

- Read where the line crosses the y-axis. No calculation needed.



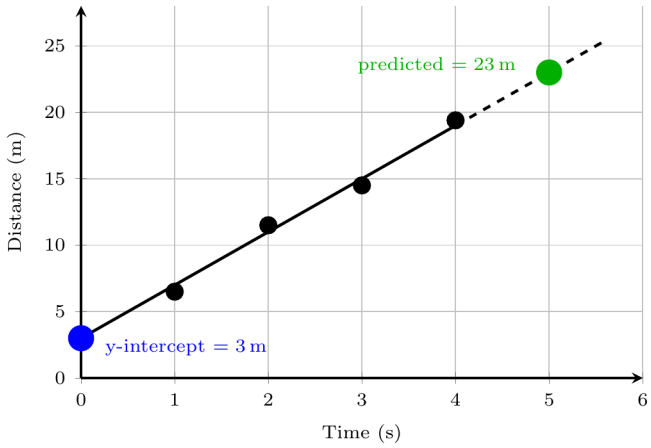
Step 4: Calculate the Slope

- Choose two points on the line that can be read from the graph.
- slope $= \frac{\text{rise}}{\text{run}} = \frac{19\ \text{m} - 3\ \text{m}}{4\ \text{s} - 0\ \text{s}} = 4 \frac{\text{m}}{\text{s}}$



Step 5: Extrapolate

- Extend the line past the last point and read the value at the target time.
- At $t = 5$ s, the line reads $d = 23$ m.



Scientific Inquiry — Scientific Method & Variables

Lesson 6 of 6

Write a hypothesis, identify variables and control and experimental groups, and evaluate the hypothesis after the experiment.

The Scientific Method

> **Important**
> The method runs: **Observation → Question → Hypothesis → Experiment → Conclusion.**

Step	What happens
Observation	Spot patterns, trends, or anomalies in data.
Question	Ask a specific, testable question based on observations.
Hypothesis	Offer a testable possible explanation.
Experiment	Run a controlled test, varying one factor at a time and measuring outcomes.
Conclusion	Analyze results to see if they support the hypothesis, contradict it, or prompt new questions.

Hypothesis

> **Note**
> A **hypothesis** is a **testable** and **falsifiable** prediction based on observations — **not** a random guess. Write it as an "**if... then...**" statement (e.g. *If* the ramp is longer, *then* the ball will roll farther).

Variables

- Independent Variable: The variable you change on purpose. Plotted on the x-axis.
- Dependent Variable: The variable you measure. Plotted on the y-axis.
- Control Variable: A variable kept constant.
- Control Group: The group that does not receive the treatment; used as a baseline for comparison.
- Experimental Group: The group that receives the treatment or independent variable.

Why Control Variables?

To run a **fair** and **repeatable** test, keep every other variable the same:

- Only the **independent variable** changes, so a change in the **dependent variable** can be traced to that one change.
- If several things changed at once, you could not tell which one caused the result.
- Another person in a different lab can repeat the test and get the same result.

Example: A student drops a ball from different heights and measures how high it bounces.

Step	In this case
Question	Does the drop height affect how high the ball bounces?
Hypothesis	If the ball is dropped from a greater height, then it will bounce higher.
Independent variable	drop height (what we change)
Dependent variable	bounce height (what we measure)
Control variable	the **same ball** (kept the same)
Experimental group	the drop from a **higher** height (gets the change)
Control group	the drop from the **original** height (baseline)

After the Experiment

- Compare data against your hypothesis.
- Was the hypothesis supported or not supported?
- A conclusion states what the evidence showed — even if the hypothesis was wrong.

Summary

- The **scientific method**: Observation → Question → Hypothesis → Experiment → Conclusion.
- A **hypothesis** is a testable "if... then..." prediction — based on observations, **not** a random guess.
- **Independent variable** = what you change; **dependent variable** = what you measure; **control variables** = what you keep the same.
- A **control group** is the baseline; the **experimental group** gets the change.

Unit 2 — Kinematics

Kinematics — Position and Reference Frames

Lesson 1 of 4

Describe position with 1-D and 2-D coordinate systems and frames of reference, add vectors, and distinguish distance from displacement.

What is Kinematics?

- **Kinematics** is the study of **motion** — how things move.
- We describe motion using **words, numbers, equations, and graphs**.
- We will study **1-D motion** (along a straight line); we also use **2-D coordinates** to describe position and direction.

Scalars and Vectors

> Note
> **Scalar**: a quantity with **magnitude** (a number and unit) only. **Vector**: a quantity with **magnitude** and **direction**.

$x \quad \Delta x \quad v \quad a$

> Direction is what separates a scalar from a vector.

Vector Addition

> Tip
> Vectors must be on the same axis (collinear): **same** direction **add**, **opposite** directions **subtract**.

Same Direction $5\ \text{m} + 3\ \text{m} = 8\ \text{m}$

Opposite Direction $5\ \text{m} + 3\ \text{m} = 2\ \text{m}$

Kinematics Variables and Symbols

Quantity	Symbol	Units	Type
position	x	m	vector →
distance	d	m	scalar
displacement	Δx	m	vector →
time	t	s	scalar
speed	s	$\frac{\mathrm{m}}{\mathrm{s}}$	scalar
velocity	v	$\frac{\mathrm{m}}{\mathrm{s}}$	vector →
acceleration	a	$\frac{\mathrm{m}}{\mathrm{s}^2}$	vector →

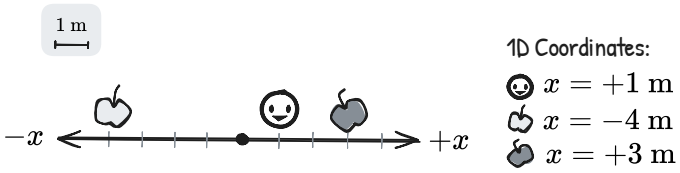
Position

Position x is the location of an object in a **coordinate system**.

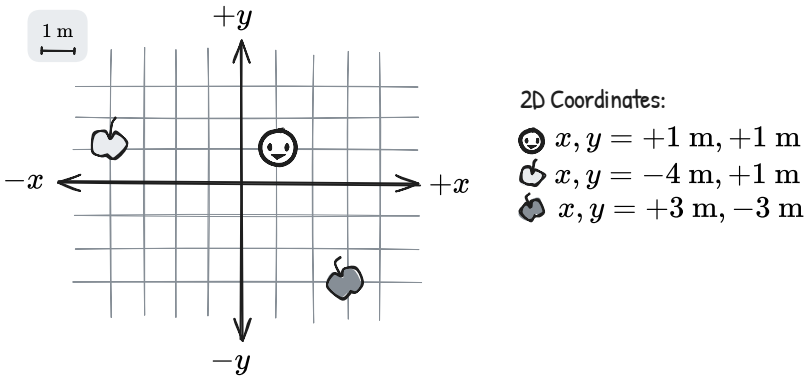
A **coordinate system** has:

- **axes** (show positive and negative directions)
- **origin** (the center of the coordinate system)
- **scale** and **units**

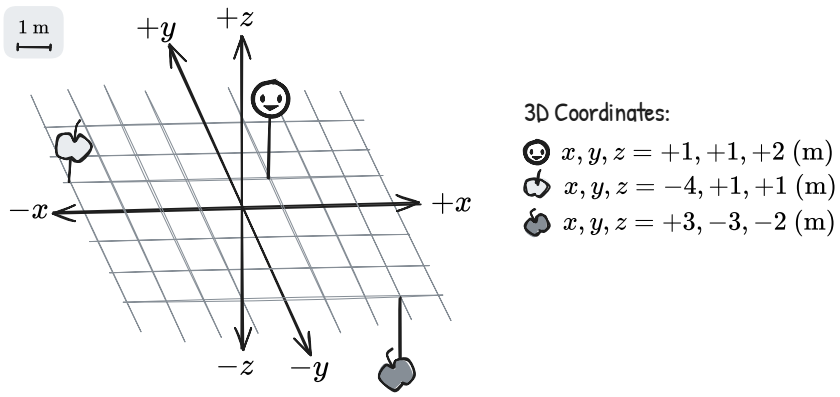
1-D Coordinate System



2-D Coordinate System



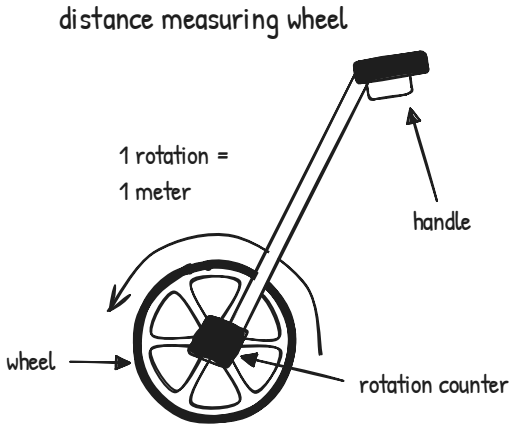
3-D Coordinate System



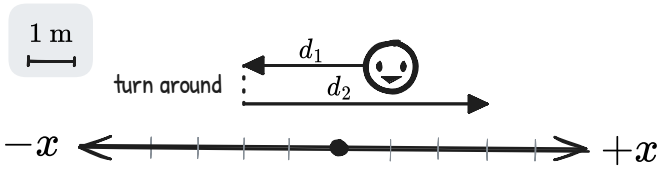
> We will only use 1-D and 2-D.

Distance

Distance measures the length of the path taken. Distance is always positive. An object's distance increases every time it changes position, regardless of direction.

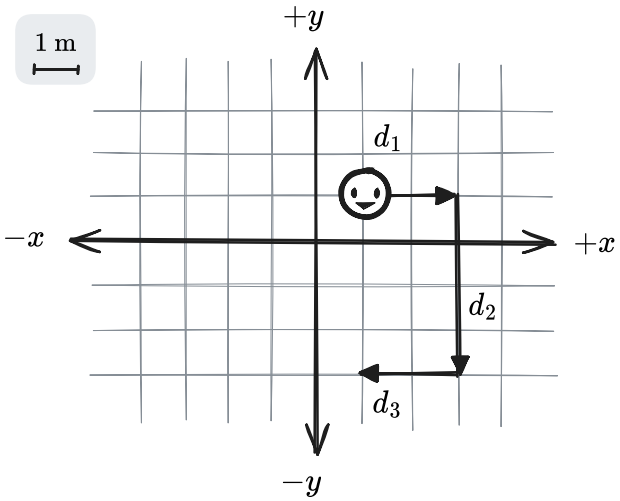


Distance in 1-D



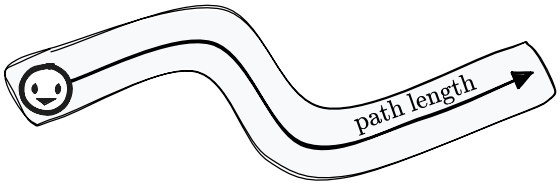
$$d = d_1 + d_2$$

Distance in 2-D



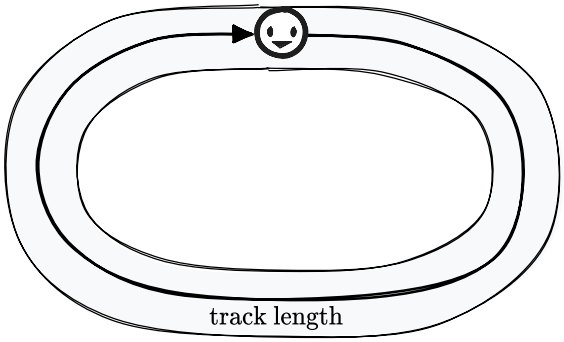
$$d = d_1 + d_2 + d_3$$

Path Distance



$$d = \text{path length}$$

Track Distance

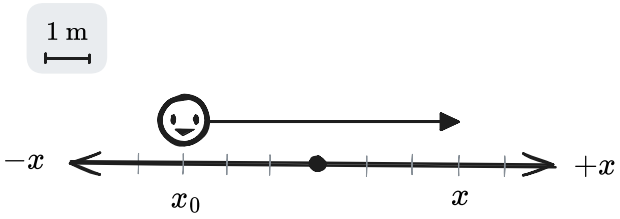


$$d = \text{number of laps} \times \text{track length}$$

Displacement

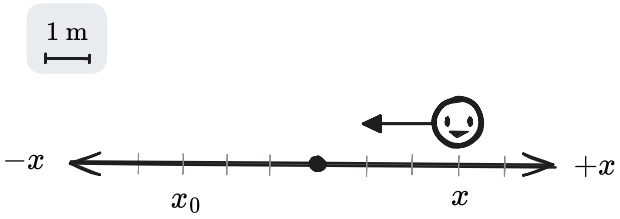
> **Important**
> **Displacement** $\Delta x = x - x_0$ is the change in position, where x is the *final position* and x_0 is the *initial position*. It is a **vector**. With several positions ($x_0, x_1, x_2, \ldots$) we can add the changes $\Delta x_1, \Delta x_2, \Delta x_3, \ldots$

1-D Displacement



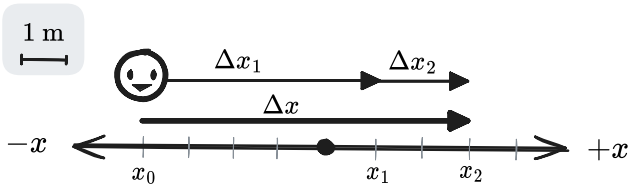
Moving toward the **right**.
Displacement is **positive**.

1-D Displacement (negative)



Moving toward the **left**.
Displacement is **negative**.

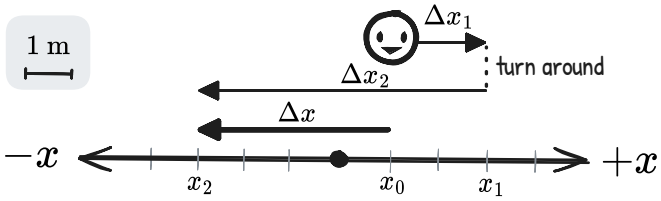
Resultant 1-D Displacement



$\Delta x = \Delta x_1 + \Delta x_2$, or...

$\Delta x = x_2 - x_0$ (final - initial)

Resultant 1-D Displacement (+/-)



$$\Delta x = \Delta x_1 + \Delta x_2 \text{ } (\Delta x_2 \text{ is negative})$$

$$\Delta x = x_2 - x_0 \text{ } (x_2 \text{ is negative})$$

Distance vs Displacement

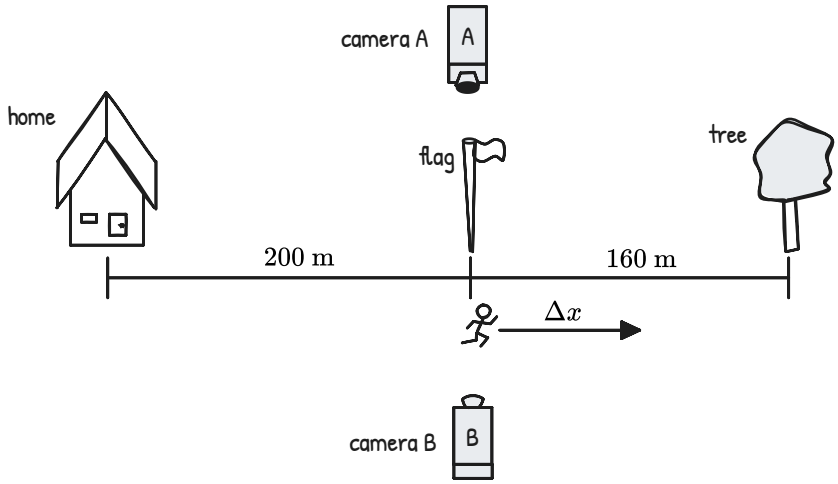
Distance	Displacement
total path length	change in position
scalar (no direction)	vector (magnitude + direction)
never negative	can be positive, negative, or zero
depends on the route taken	depends only on start and finish

Frames of Reference

> Note
> A **frame of reference** is the view from which motion is measured. Every measurement is made **relative to** a **reference point** — so the **same motion can look different** from different frames of reference.

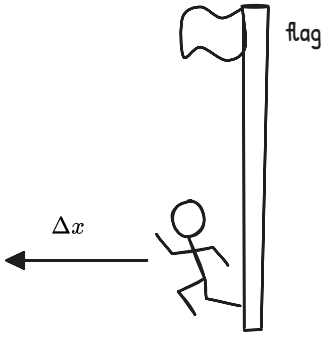
Frame of Reference Example

Camera A and Camera B record the man running in opposite directions from the flag (reference point).



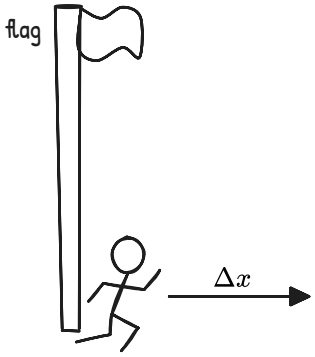
Camera A's View

In this view, the man is running to the left of the flag.



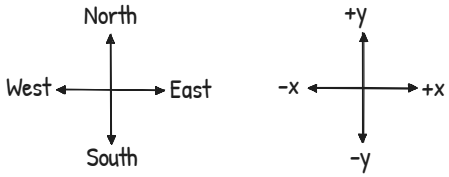
Camera B's View

In this view, the man is running to the right of the flag.



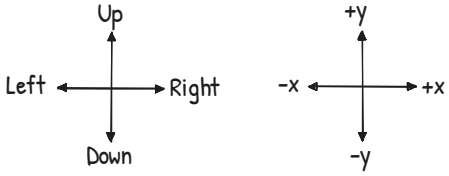
Compass Directions

Compass directions form a North-South axis and an East-West axis. The four directions correspond to positive and negative as shown:



Basic Directions

Up, down, left, right are commonly used. The four directions correspond to positive and negative as shown:



Summary

- **Scalar** = magnitude only; **vector** = magnitude + direction.
- **Position** is described using a coordinate system, which has an origin, axes, units, scale.
- **Distance** is a scalar; **displacement** is a vector.
- A **reference point** is the origin used to measure position.
- How motion is described depends on the **frame of reference**.

Kinematics — Speed and Velocity

Lesson 2 of 4

Calculate speed and velocity, and combine velocity vectors to find resultant and relative velocities.

Time

> Note
> ***Time*** (t) is the duration between two states. A *state* is the set of quantities (e.g. position, velocity) an object has at one moment.

- Δx: *A student walked from the **math classroom** to the **biology classroom** in **120 seconds**.*
- Δv: *A plane needed **30 seconds** to accelerate from **rest** to **100 m/s**.*

Kinematic equations that use time

- $\bar{v} = \dfrac{\Delta x}{t}$
- $s = \dfrac{d}{t}$
- $a = \dfrac{\Delta v}{t} = \dfrac{v - v_0}{t}$
- $\Delta x = x - x_0$

Kinematic equations that use time (Grade 7)

- $v = v_0 + at$
- $\Delta x = v_0 t + \frac{1}{2}at^2$
- $v^2 = v_0^2 + 2a\Delta x$

Speed

> Note
>
> **Speed** is a **scalar** — how fast something moves. $s = \frac{d}{t}$. e.g. the speed of the bus is $40\frac{\text{m}}{\text{s}}$.

Speed and Velocity Equations

> Note
>
> **Velocity** is a **vector** — how fast something moves *and in what direction*. $v = \frac{\Delta x}{t}$. e.g. the velocity of the bus is $40\frac{\text{m}}{\text{s}}$ east.

Problem Solving Steps

1. **What am I given? What am I finding?**
2. **What equation connects them?**
3. **Plug in and solve.**
4. **Check the units and direction (if applicable).**

Example: Speed

A train travels $2000\ \text{m}$ in $60\ \text{s}$. What is its speed?

1. $d = 2000\ \text{m}$, $t = 60\ \text{s}$, $s = ?$
2. $s = \frac{d}{t}$
3. $s = \frac{2000\ \text{m}}{60\ \text{s}} = 33.3\ \frac{\text{m}}{\text{s}}$
4. $\frac{m}{s}$✓ scalar✓

Example: Distance

A dog runs at a constant speed of $5\frac{m}{s}$ for a time of $30\ \text{s}$. What is the distance traveled?

1. $s = 5\frac{m}{s}$, $t = 30\ \text{s}$, $d = ?$
2. $d = st$
3. $d = 5\frac{m}{s} \cdot 30\ \text{s} = 150\ \text{m}$
4. m✓ scalar✓

Example: Velocity

A bird flies $500\ \text{m}$ east in $25\ \text{s}$. What is its velocity?

1. $\Delta x = 500\ \text{m}$ east, $t = 25\ \text{s}$, $v = ?$
2. $v = \frac{\Delta x}{t}$
3. $v = \frac{500\ \text{m}}{25\ \text{s}} = 20\ \frac{\text{m}}{\text{s}}$ east
4. $\frac{\text{m}}{\text{s}}$✓ east✓

Example: Displacement

A balloon floats upward at a constant velocity of $2.5\frac{\text{m}}{\text{s}}$ for a time of $20\ \text{s}$. What is its displacement?

1. $v = 2.5\frac{\text{m}}{\text{s}}$, $t = 20\ \text{s}$, $\Delta x = ?$
2. $\Delta x = vt$

3. $\Delta x = 2.5\frac{\text{m}}{\text{s}} \cdot 20\ \text{s} = 50\ \text{m up}$

4. $\frac{\text{m}}{\text{s}}$✓ up✓

Resultant Velocity

> Tip
> **Resultant velocity** is the single velocity you get when you combine two or more velocity vectors acting on the **same object**. Add same-direction vectors, subtract opposite ones.

Resultant Velocity (Same Direction)

A boat moves $3\frac{\text{m}}{\text{s}}$ east while the current pushes $1\frac{\text{m}}{\text{s}}$ east.

$v_{\text{resultant}} = 3\frac{\text{m}}{\text{s}} + 1\frac{\text{m}}{\text{s}} = 4\ \frac{\text{m}}{\text{s}}\ \text{east}$

Resultant Velocity (Opposite Direction)

A boat moves $3\frac{\text{m}}{\text{s}}$ east while the current pushes $1\frac{\text{m}}{\text{s}}$ west.

$v_{\text{resultant}} = 3\frac{\text{m}}{\text{s}} - 1\frac{\text{m}}{\text{s}} = 2\frac{\text{m}}{\text{s}}\ \text{east}$

Relative Velocity

> Note
> **Relative velocity** is the velocity of one object measured from the **frame of reference** of another object: $v_{BA} = v_B - v_A$.

Relative Velocity (Same Direction)

Two ships move east: ship A at $10\ \text{m/s}$ and ship B at $15\ \text{m/s}$.

Velocity of B relative to A:

- $v_{BA} = v_B - v_A = 15\frac{\text{m}}{\text{s}} - 10\frac{\text{m}}{\text{s}} = 5\frac{\text{m}}{\text{s}}$ east.

Velocity of A relative to B:

- $v_{AB} = v_A - v_B = 10\ \frac{\text{m}}{\text{s}} - 15\ \frac{\text{m}}{\text{s}} = 5\frac{\text{m}}{\text{s}}$ west.

Summary

- **Speed** = how fast (scalar); **velocity** = how fast and which way (vector).
- $s = \dfrac{d}{t}$ and $v = \dfrac{\Delta x}{t}$
- Units of speed and velocity: $\frac{\text{m}}{\text{s}}$.

Summary (Grade 7)

- **Resultant velocity** is the sum of two velocities on the object. $v_{resultant} = v_1 + v_2$
- **Relative velocity** is the difference of velocity between two objects. $v_{BA} = v_B - v_A$ (How B sees A)

Kinematics — Speed, Velocity, and Acceleration

Lesson 3 of 4

Distinguish constant, average, and instantaneous speed and velocity, and describe acceleration as speeding up, slowing down, or changing direction.

Average Speed

> Note
> Average speed is total distance divided by total time: $s_{\text{average}} = \dfrac{d_{\text{total}}}{t_{\text{total}}}$.

Average Speed

A van travels **120 m east** in 10 s, then **80 m east** in 5 s, then **100 m west** in 5 s. What is its average speed?

- $d_{total} = 120\ \text{m} + 80\ \text{m} + 100\ \text{m} = 300\ \text{m}$

- $t_{total} = 10\ \text{s} + 5\ \text{s} + 5\ \text{s} = 20\ \text{s}$
- $s_{average} = ?$
- $s_{\text{average}} = \frac{d_{\text{total}}}{t_{\text{total}}} = \frac{300\ \text{m}}{20\ \text{s}} = 15\ \frac{\text{m}}{\text{s}}$
- $\frac{\text{m}}{\text{s}}$, scalar

Average Velocity

> Note
>
> Average velocity is total displacement divided by total time: $v_{\text{average}} = \dfrac{\Delta x_{\text{total}}}{t_{\text{total}}}$.

Average Velocity

A van travels **120 m east** in 10 s, then **80 m east** in 5 s, then **100 m west** in 5 s. What is its average velocity?

- $\Delta x_{total} = 120\ \text{m} + 80\ \text{m} - 100\ \text{m} = 100\ \text{m east}$
- $t_{total} = 20\ \text{s}$
- $v_{average} = ?$
- $v_{average} = \frac{\Delta x_{\text{total}}}{t_{\text{total}}} = \frac{100\ \text{m}}{20\ \text{s}} = 5\ \frac{\text{m}}{\text{s}}\ \text{east}$
- $\frac{\text{m}}{\text{s}}$, east

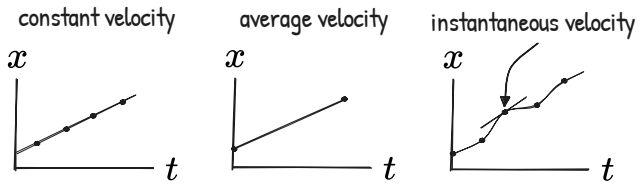
Constant Speed and Constant Velocity

- **Constant speed**: the rate of distance traveled never changes.
- **Constant velocity**: the **magnitude and direction** of velocity never change. This only happens when moving in a **straight line** at a **constant speed**.
- A constant speed but changing direction is **not** constant velocity.

Instantaneous Speed and Velocity

> Note
>
> On a trip, speed or direction can change from moment to moment. **Instantaneous speed** is the speed at one exact instant; **instantaneous velocity** is the velocity (speed **and** direction) at one instant.

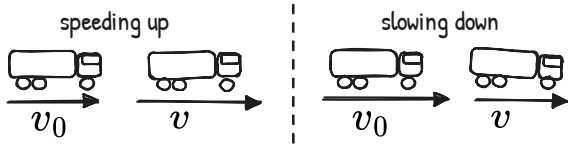
Constant vs Average vs Instantaneous



- slope = velocity
- constant velocity has a constant slope
- average velocity the slope between the final and initial positions
- instantaneous velocity may be different at every time

Speeding Up vs Slowing Down

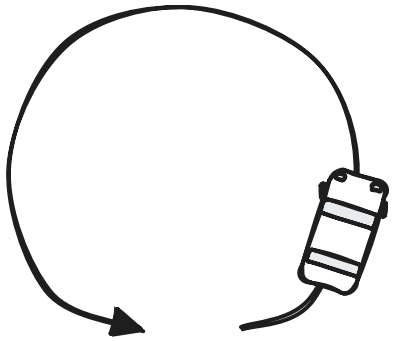
- A **change in speed** means the object covers distance at a different rate.
- In vector terms, this is a change in the **magnitude** of the velocity.



A Change in Direction is a Change in Velocity

- Velocity is a **vector**: it has magnitude **and** direction.
- Changing **either one** changes the velocity.

- A car going around a roundabout at a constant speed does *not* have a constant velocity:



What is Acceleration?

> Important
> **Acceleration** is the **rate at which velocity changes** — a **vector**. An object accelerates when it **speeds up**, **slows down**, or **changes direction**.

Acceleration Equation and Units

> Tip
> $a = \dfrac{\Delta v}{t} = \dfrac{v - v_0}{t}$
> - v_0 = initial velocity, v = final velocity, t = time; $\Delta v = v - v_0$ is the **change in velocity**.
> - Units: $[a] = \dfrac{\text{m}}{\text{s}^2}$ — read "metres per second squared."

Summary

- **Average speed** uses total **distance**; **average velocity** uses total **displacement**.
- **Constant** means never changes; **average** is for the whole trip.
- An object accelerates when it **speeds up**, **slows down**, or **changes direction**.
- **Acceleration**: $a = \dfrac{v - v_0}{t}$

Summary (Grade 7)

- **Instantaneous** speed or velocity is the value at one exact moment.

Kinematics — Motion Graphs

Lesson 4 of 5

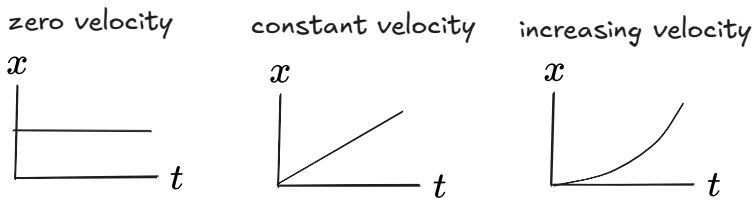
Read position-time and velocity-time graphs, using slopes to find velocity and acceleration.

Motion Graphs

The motion of an object can be shown on a graph with **time** on the x-axis:

- **Position-time graph** — how position changes with time
- **Velocity-time graph** — how velocity changes with time

Position-Time Graphs



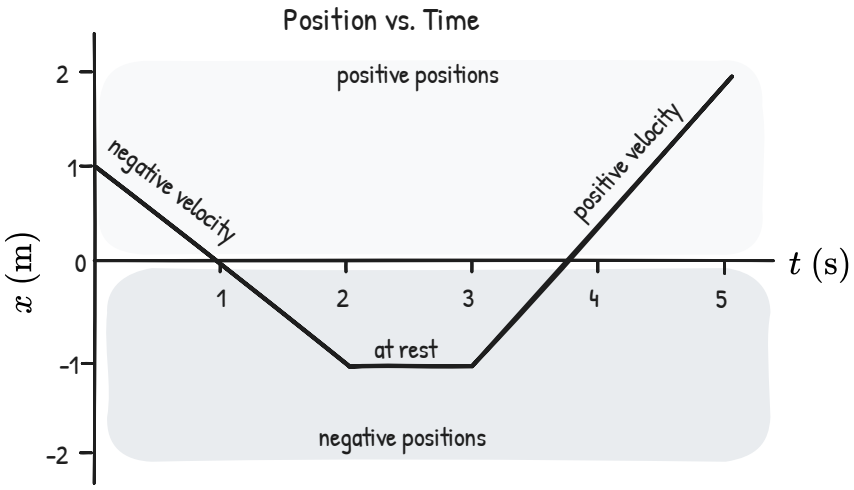
> Tip
> On a position-time graph, the y-**value** is the **position** and the **slope** is the **velocity**.

Velocity from a Position-Time Graph

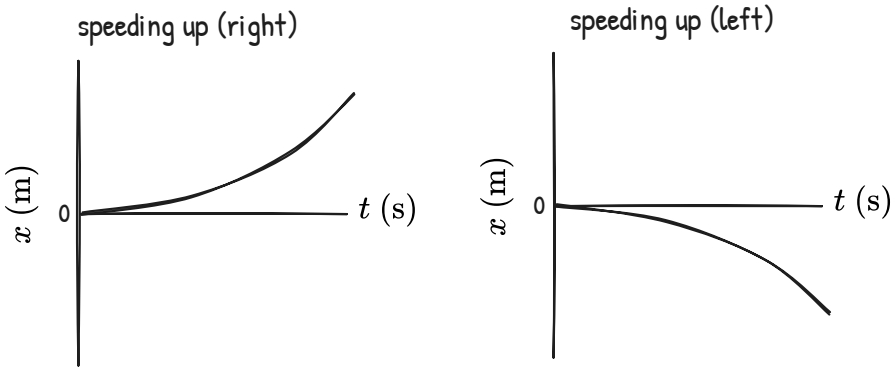
> **Important**
>
> On a **position-time** graph, the **slope** is the **velocity**: $v = \frac{\Delta x}{t} = \frac{x - x_0}{t}$ (rise over run).

- A steeper slope means a **greater velocity**.
- Positive slope: $v \rightarrow$ or $v \uparrow$
- Negative slope: $v \leftarrow$ or $v \downarrow$

Features of a Position-Time Graph

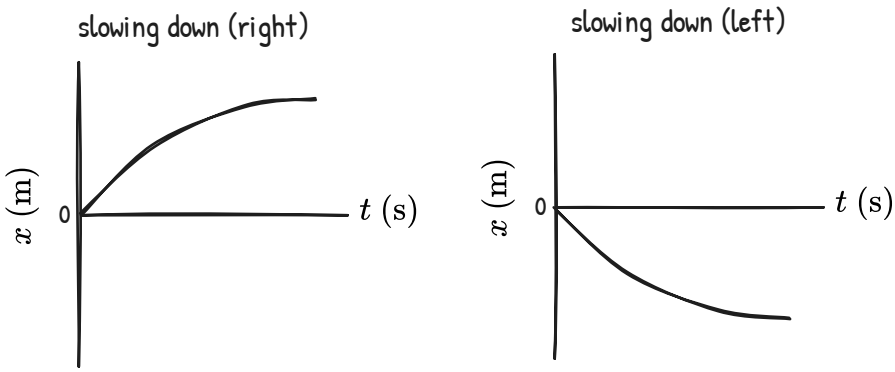


Position-Time: Speeding Up



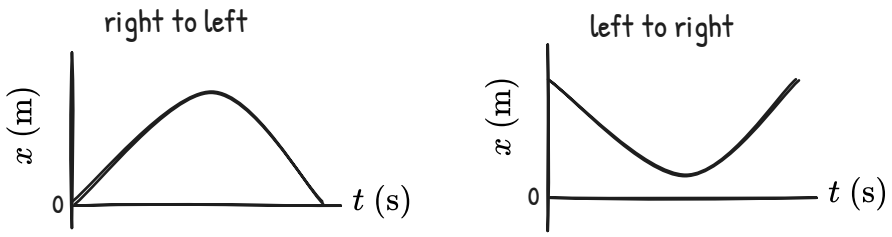
> **Tip**
>
> Speeding up: the slope gets steeper.

Position-Time: Slowing Down



> **Tip**
> Slowing down: the slope gets flatter.

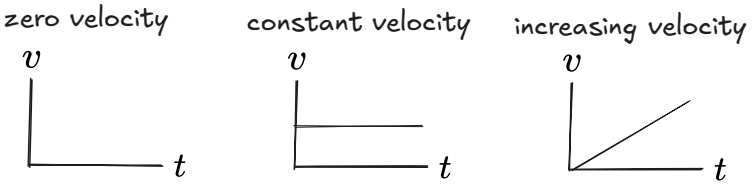
Position-Time: Changing Direction



> **Warning**
> Changing direction: the slope changes **positive to negative** (or the reverse). The position does not need to be positive.

*This can happen anywhere. Position does not need to be positive.

Velocity vs Time



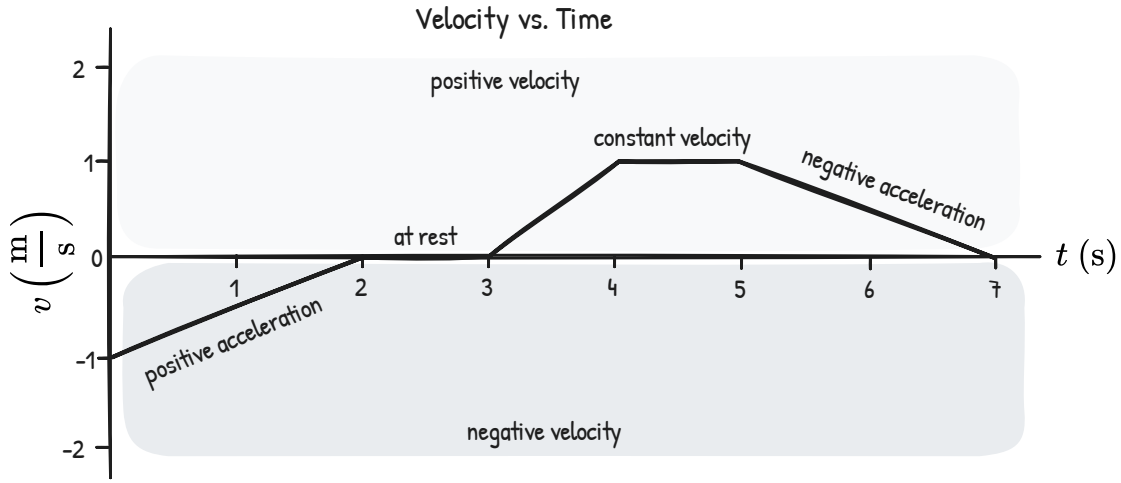
> **Tip**
> On a velocity-time graph, the y-**value** is the **velocity** and the **slope** is the **acceleration**.

Acceleration from a Velocity-Time Graph

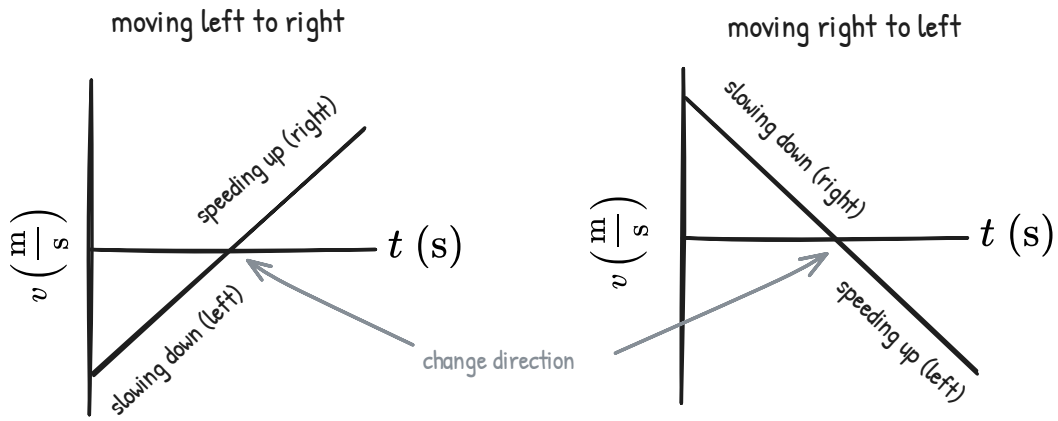
> **Important**
> On a **velocity-time** graph, the **slope** is the **acceleration**: $a = \frac{\Delta v}{t} = \frac{v - v_0}{t}$ (rise over run).

- A steeper slope means a **greater acceleration**.
- Positive slope: $a \rightarrow$ or $a \uparrow$
- Negative slope: $a \leftarrow$ or $a \downarrow$

Features of a Velocity-Time Graph



Velocity-Time: Changing Speed and Direction



Summary

- **Position-time** graphs show position
  - slope = velocity
- **Velocity-time** graphs show velocity
  - slope = acceleration

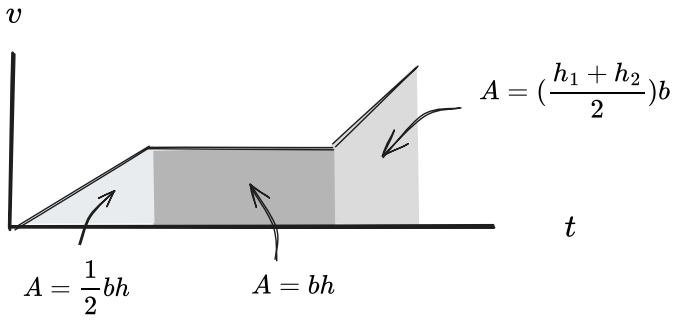
Kinematics — Area Under a Velocity-Time Graph

Lesson 5 of 5

Find displacement from the area under a velocity-time graph, including its units and sign.

Area Under the Curve

> **Important**
> The **area** between the line on a **velocity-time** graph and the horizontal axis equals the **displacement**.



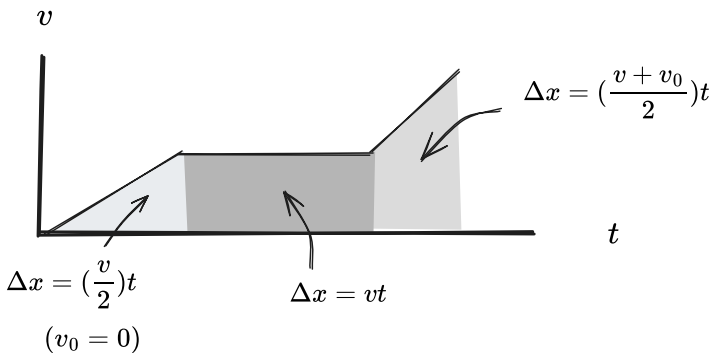
Units of Area Under the Curve

$$\text{area} = \text{velocity} \cdot \text{time} = \frac{\text{m}}{\text{s}} \cdot \text{s} = \text{m}$$

- The unit are **meter** which is the unit of **displacement**.

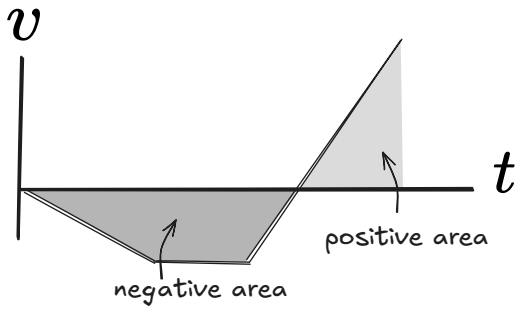
Area Under the Curve with Kinematic Equations

- Each shape corresponds to a kinematic equation. Notice the similarity between the triangle and trapezoid.



Positive and Negative Displacement

> ⚠ Warning
> Area **above** the axis is **positive** displacement; area **below** the axis is **negative** displacement. Add them to get the total displacement.



Summary

- The **area** under a **velocity-time** graph is the **displacement**.
- Area **above** the axis is **positive** displacement; area **below** is **negative**.
- Units of area: $\frac{\mathrm{m}}{\mathrm{s}} \cdot \mathrm{s} = \mathrm{m}$.